

Alexandru D. Cozma

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EDUCATION

Florida Polytechnic University

Expected May 2028

Bachelor's in Engineering Physics & Applied Mathematics, Focus in Electrical Engineering

Lakeland, FL

Activities: Society of Physics Students (SPS), IEEE HKN Mu Omega Chapter, Rock Climbing Club

Relevant Coursework: *Math:* Linear Algebra, Differential Equations, Applied PDEs, Complex Analysis.

Physics: Classical Mechanics, Quantum Mechanics I, Electricity & Magnetism I, Modern Physics.

Lab & Computational: Experimental Techniques in Physics, Computational Physics I.

Engineering & CS: Circuits I, Materials Science, Object-Oriented Programming (C++).

Bergen Community College

Graduated May 2025

Associate in Engineering Science

Paramus, NJ

Activities: 3SP Research Intern & Volunteer, Math Club, STEM Student Union, Computer Science Club

EXPERIENCE

Owner / Operator

May – Nov 2024

Cozma Landscaping (side business)

Franklin Lakes, NJ

- Ran a seasonal residential landscaping business with a rotating 5-person crew, serving 3 regular clients and roughly a dozen one-offs across mowing, mulching, powerwashing, seasonal cleanups, and small design/rebedding jobs.
- Handled all client-facing operations end-to-end: quoting, scheduling, expense/revenue tracking, and payroll, netting ~\$5,000 in revenue over the season.
- Learned to price work accurately after under-quoting an early job that ran over scope; built a repeatable quote template and pre-visit research routine that made estimates more reliable.

Research Intern — STEM Student Scholars Program (3SP)

Apr – Aug 2025

Bergen Community College

Paramus, NJ

- Team member on ARA (Anthropomorphic Robotic Arm), a multi-year project developing an affordable 3D-printed prosthetic hand; owned the design of the power and charging system.
- Specified a custom 4S Li-ion pack (~14.8 V) with dedicated BMS for cell balancing and protection, and USB-C fast charging, targeting a ~\$120 total power-system BOM vs. off-the-shelf alternatives.
- Presented the design in the team's final research symposium; the pack was not built during the internship due to component procurement constraints outside the team's control.

TECHNICAL SKILLS

Programming: C/C++, Python, MATLAB, Bash/Powershell

Hardware: Arduino, PCB Design (Altium), 3D Printing, Soldering, Electronics Prototyping

Software: Fusion 360, Cura, Altium Designer, Linux, Git, L^AT_EX

Certifications: Heartsaver CPR/AED (AHA)

PROJECTS

Cosmic-Ray Muon Detector (Tomography Prototype) – *Primary Electronics Designer*

Apr – May 2026

- Designed the analog signal chain and microcontroller readout for a cosmic-ray muon detector—a proof-of-concept for muon tomography, which images dense objects using high-energy particles from space.
- Validated the completed detector by observing muon events at ~1 per 3–4 s, matching the predicted rate for the scintillator's 58 cm² collection area.

NJ Highlands Lookout Atlas – *Solo Project*

May 2026 – Current

- Built a Python pipeline processing 140M pixels of USGS elevation data (~4,300 mi² across NJ/NY/PA) to rank scenic overlooks across three models—panoramic openness, NYC skyline sightlines, and water proximity.
- Produced a self-contained interactive web atlas of 45 ranked lookouts with toggleable map layers, distributed as a single HTML file with no backend.